

Multiple Choice (10 marks)

1. Which of the following is NOT supervised learning?
 - (a) PCA
 - (b) Decision Tree
 - (c) Linear Regression
 - (d) Naive Bayesian
2. What would you do in PCA to get the same projection as SVD?
 - (a) Transform data to zero mean
 - (b) Transform data to zero median
 - (c) Not possible
 - (d) None of these
3. Statement 1| As of 2020, some models attain greater than 98% accuracy on CIFAR-10. Statement 2| The original ResNets were not optimized with the Adam optimizer.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True
4. Existential risks posed by AI are most commonly associated with which of the following professors?
 - (a) Nando de Freitas
 - (b) Yann LeCun
 - (c) Stuart Russell
 - (d) Jitendra Malik
5. Statement 1| ImageNet has images of various resolutions. Statement 2| Caltech-101 has more images than ImageNet.
 - (a) True, True
 - (b) False, False
 - (c) True, False
 - (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: The Penn Treebank is primarily focused on image classification tasks rather than language modeling in natural language processing.
7. True or False: Fully connected networks are widely regarded as the best choice for high-resolution image classification tasks.
8. True or False: K-fold cross-validation is quadratic in K, suggesting that the computation time grows with the square of the number of folds.
9. True or False: Maximum Likelihood Estimation (MLE) can produce estimates that have high variance, making them less desirable in practice.
10. True or False: Dropout randomly sets all activation values to zero during training, effectively disabling the entire layer.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to check whether two given lines are parallel or not.
12. Write a function to find the previous palindrome of a specified number.

End of Paper